

ELEC 221 Lecture 03

DT impulse response and convolution sum; CT convolution integral

Thursday 17 September 2026

Announcements

- Class back in person
- Quiz 2 on Tuesday
- Monday tutorial: tutorial assignment 1 (intro to Python)

Last time

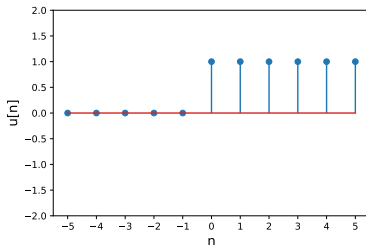
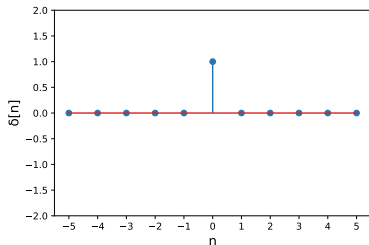
We defined **LTI (linear, time-invariant) systems**.

Linearity:

Time invariance:

Last time

We defined the DT unit impulse and unit step



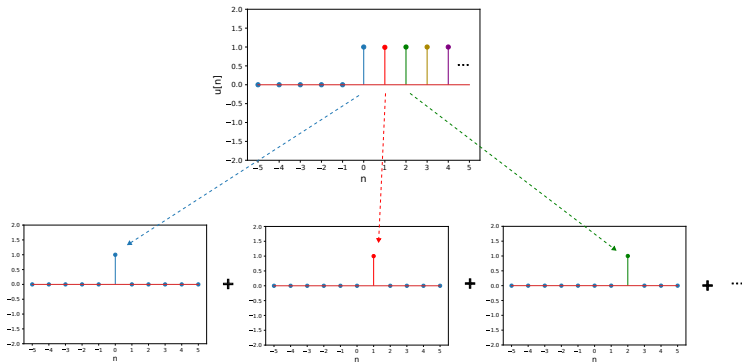
We expressed them in terms of each other.

Today

Learning outcomes:

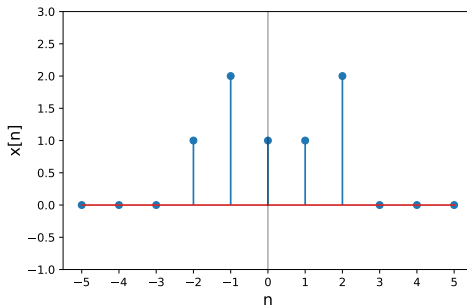
- Define the DT impulse response
- Define the convolution sum and use it to compute the output of a system
- Define the CT unit impulse and step functions
- Define the convolution integral and use it to compute the output of a system

Signals as superpositions of weighted, shifted impulses



The unit impulse as a sampler

Multiplying by a shifted impulse “samples” the signal at that point:



The impulse response

How does an LTI system respond to a signal

Suppose it sends $\delta[n - k] \rightarrow h_k[n]$

$h_k[n]$ is called the **impulse response**.

Real-world example: nerve conduction study

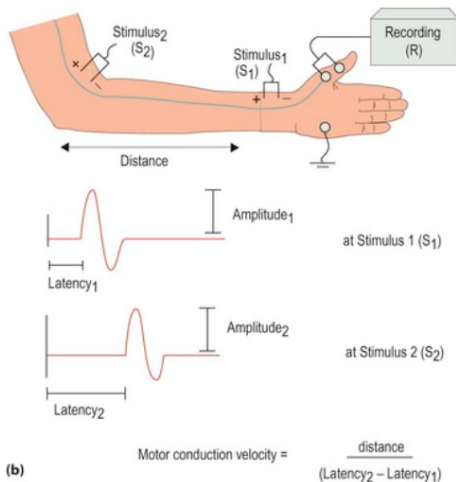


Image source: <https://neupsykey.com/nerve-conduction-studies-and-electromyography/>

The impulse response and time-invariance

What if the system is also time invariant?

Then

The convolution sum

If we know how a **linear** system responds to the unit impulse, we can learn how it responds to **any other signal**!

This is the **convolution sum**. We are “convolving” the sequences $x[n]$ and $h[n]$.

Exercise: impulse response

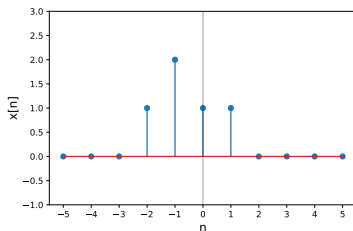
Consider an LTI system with input/output relationship

$$y[n] = 2x[n] + x[n - 1]$$

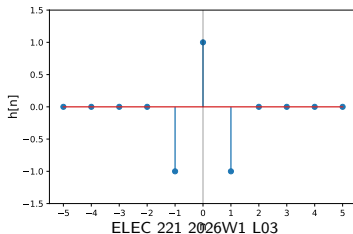
What is the impulse response of the system?

Example: convolution sum

Consider the signal

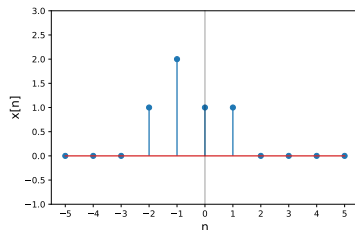


input to a system with impulse response



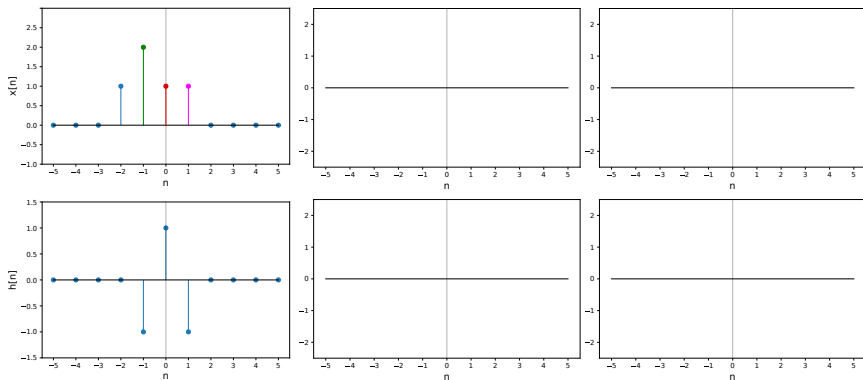
Example: convolution sum

To learn the system output, we must consider the contribution of each weighted impulse response:

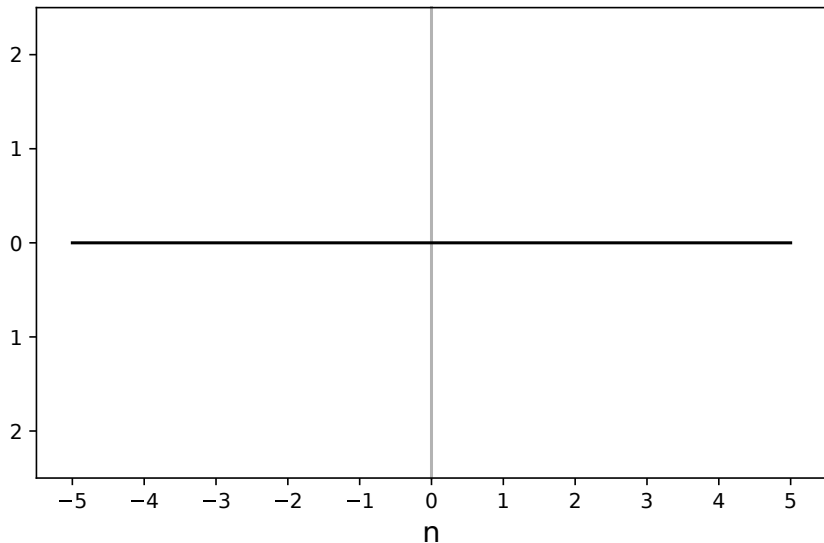


Only $x[k] \neq 0$ only for $k \in \{-2, -1, 0, 1\}$. So need to determine $x[k]h[n-k]$ for these cases, and sum them.

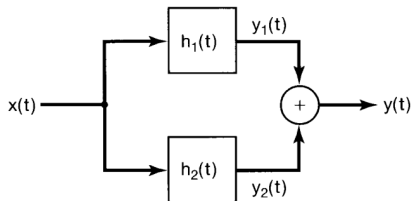
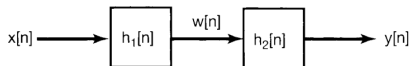
Example: convolution sum



Example: convolution sum



Properties of convolutions



Convolution is:

- Associative:

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

- Commutative:

$$x[n] * h[n] = h[n] * x[n]$$

- Distributive:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

Image credits: Signals and Systems 2nd ed., Oppenheim

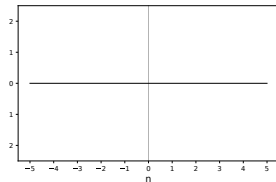
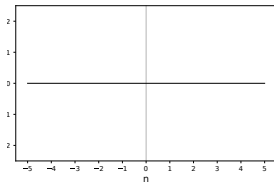
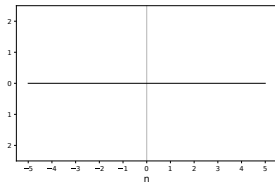
Example: convolution sum

Consider an LTI system with impulse response

$$h[n] = 3\delta[n] + 2\delta[n+1]$$

What is output of the system if

$$x[n] = \left(\frac{2}{3}\right)^n u[n]$$



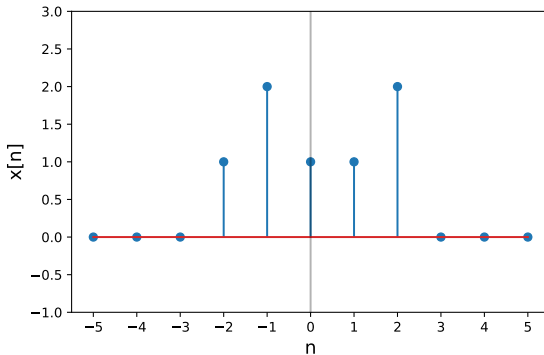
Example/exercise: convolution sum

What is output of the system

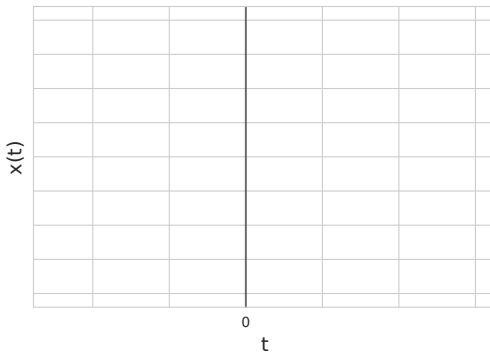
$$x[n] = \left(\frac{2}{3}\right)^n u[n], \quad h[n] = 3\delta[n] + 2\delta[n+1]$$

Shifted, weighted impulses

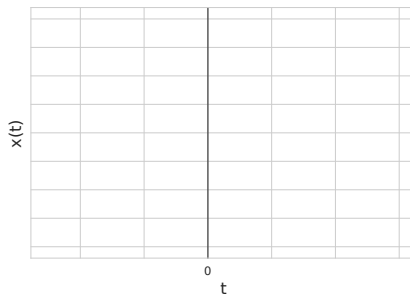
Return to this picture:



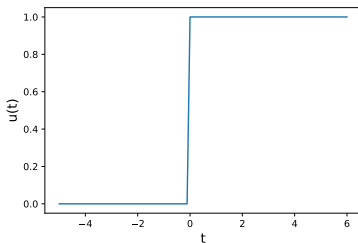
Shifted, weighted impulses



The CT unit impulse



The CT unit step



Just like in DT, the unit impulse and step are related:

The convolution integral

The CT analogue of convolution sum is the **convolution integral**.

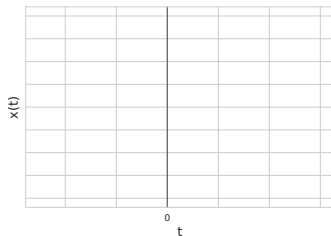
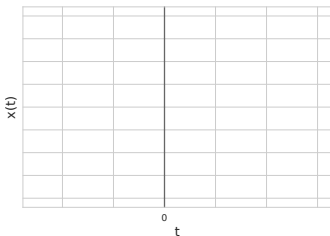
where $h(t)$ is the **CT impulse response**.

It has the same properties (commutative, associative, distributive).

Example: convolution

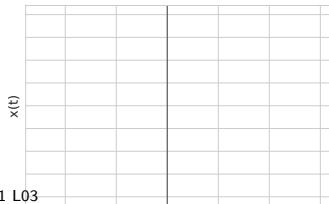
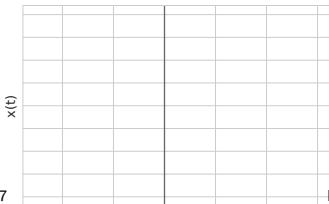
(Oppenheim Ex. 2.6 Var.) Consider system with impulse response

What is the output of the system for the input signal



Example: convolution

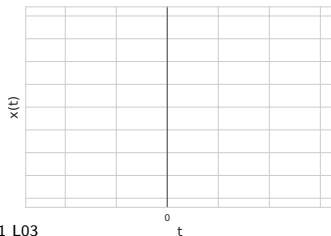
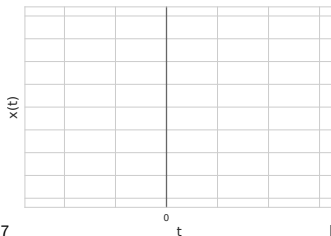
Example: convolution



Exercise: convolution

(Oppenheim 2.8) Consider system with impulse response

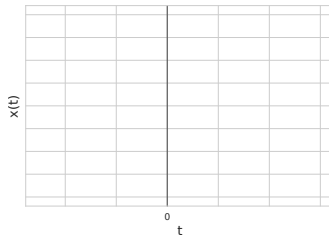
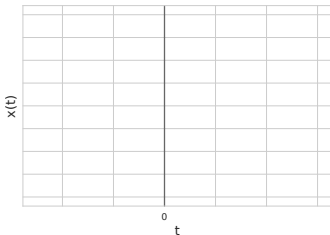
What is the output of the system for the input signal



Exercise: convolution

Direct integration:

Visual intuition:



Impulse response and analysis of LTI systems

To reiterate: the convolution sum

and convolution integral

show that as long as we know how a system responds to a unit impulse, we can determine its response to any other signal.

Knowledge of the impulse response also allows us to determine key properties of systems such as causality, invertibility, and stability.

Impulse response and memory

A system is memoryless if the output depends only on the input at the same time. This implies $h[n] = 0$ for $n \neq 0$, meaning

(And analogous for CT case)

Impulse response and invertibility

If a system is invertible, it has an inverse system.
Suppose impulse response of a system is $h(t)$. Then

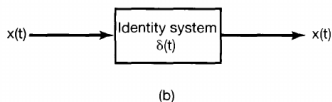
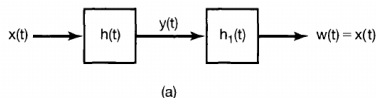


Figure 2.26 Concept of an inverse system for continuous-time LTI systems. The system with impulse response $h_1(t)$ is the inverse of the system with impulse response $h(t)$ if $h(t) * h_1(t) = \delta(t)$.

(And analogous for DT case. We will see this later in the course.)

Impulse response and stability

Suppose we have a signal $x(t)$ with a bounded input, $|x(t)| \leq B$.
If the system is stable, the output should be bounded.

Impulse response and stability

As long as

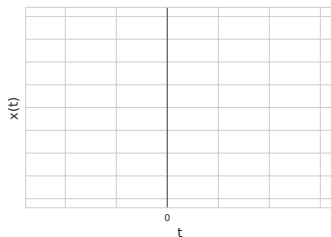
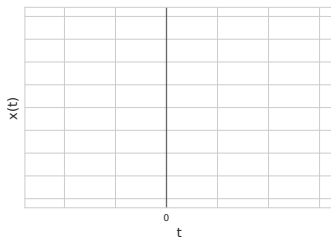
is bounded (i.e., $h(t)$ is absolutely integrable), then the system will be stable.

(And analogous for DT case)

Example/exercise: stability

Consider systems A and B with impulse responses

Are they stable?



Example/exercise: stability

Impulse response and causality

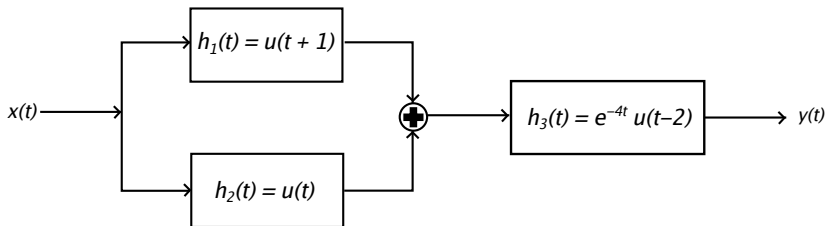
Recall definition of causal signal and consider the convolution sum:

What properties does $h[n]$ need to have for system to be causal?

(Analogous holds for CT systems)

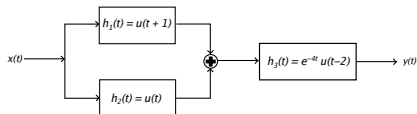
Final exercise

Consider the following combination of systems:



Is this system causal and/or stable?

Final exercise



For next time

Content:

- (Complex) exponential signals
- The continuous-time Fourier series representation

Action items:

- ① Study for quiz 2 on Tuesday
- ② Check PrairieLearn for practice assignment problems

Recommended reading:

- From today's class: Oppenheim 1.4, 2.1-2.3
- practice problems: 2.1-2.12, 2.14-16, 2.21, 2.22, 2.28, 2.29
- For next class: Oppenheim 1.3, 3.0-3.3